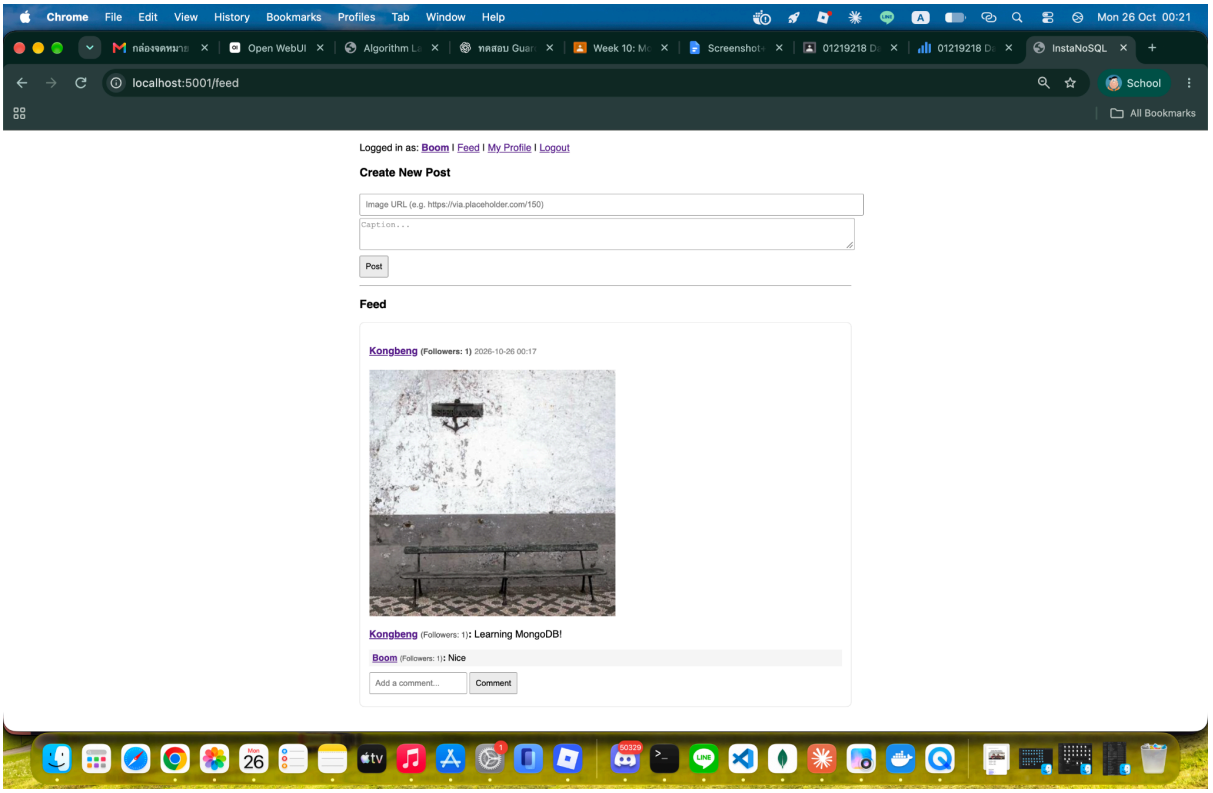
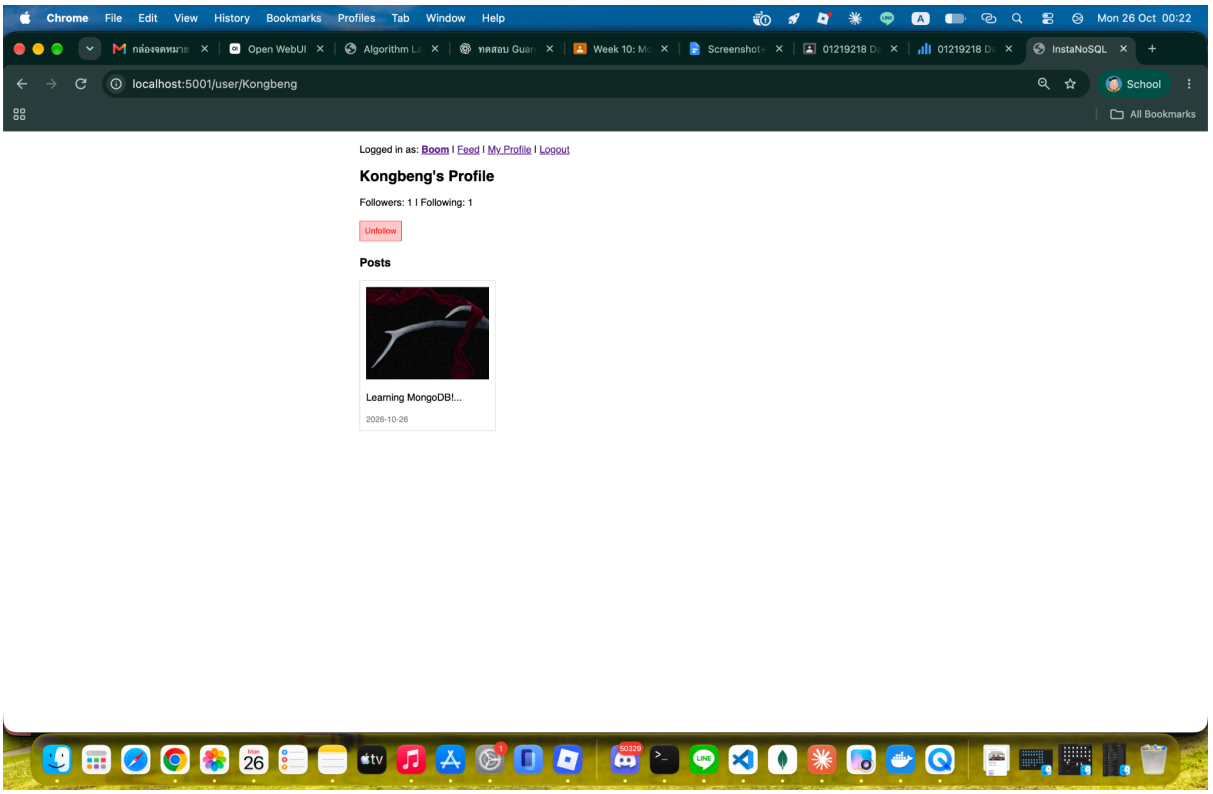
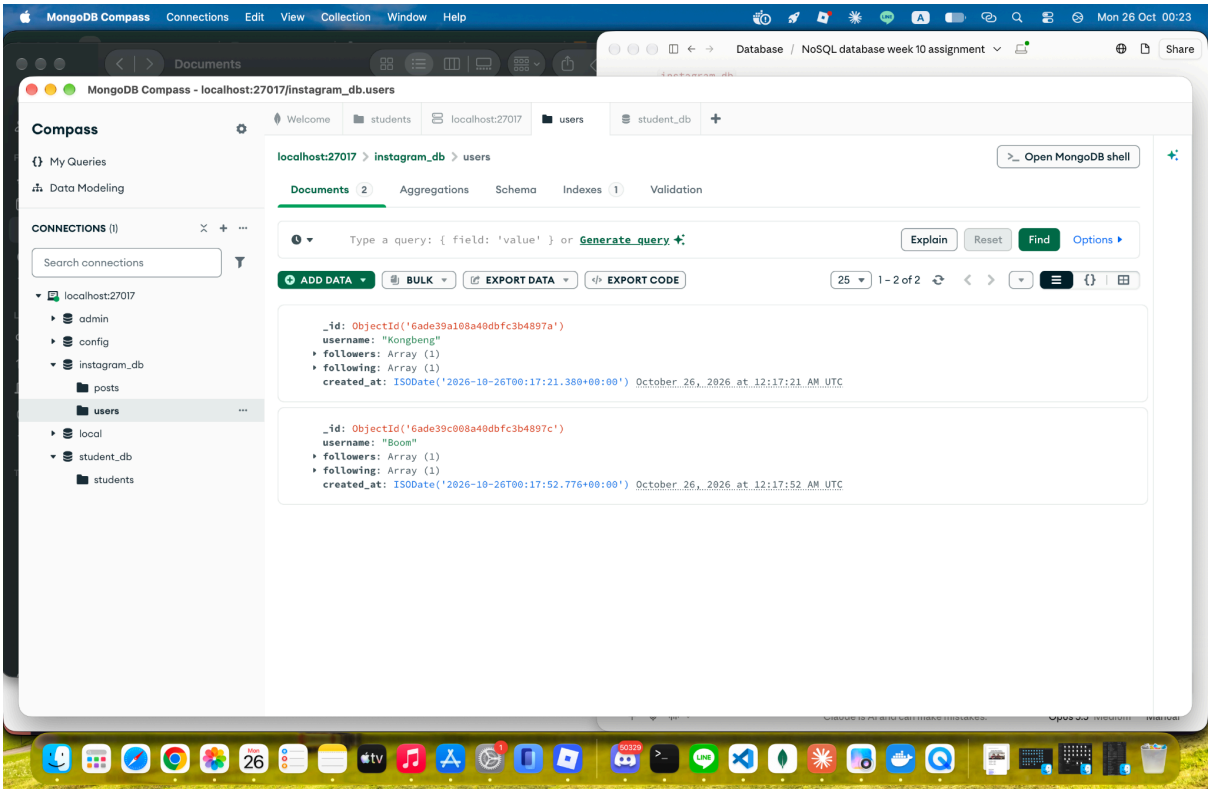
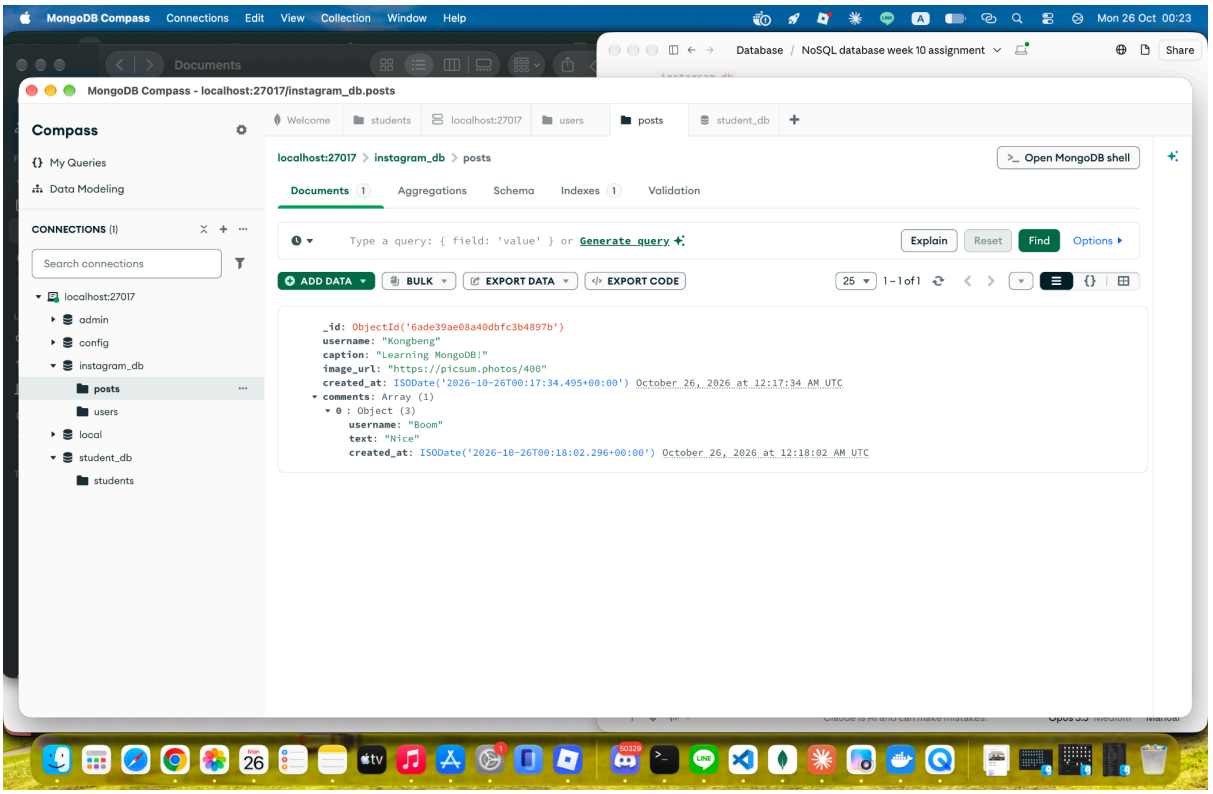
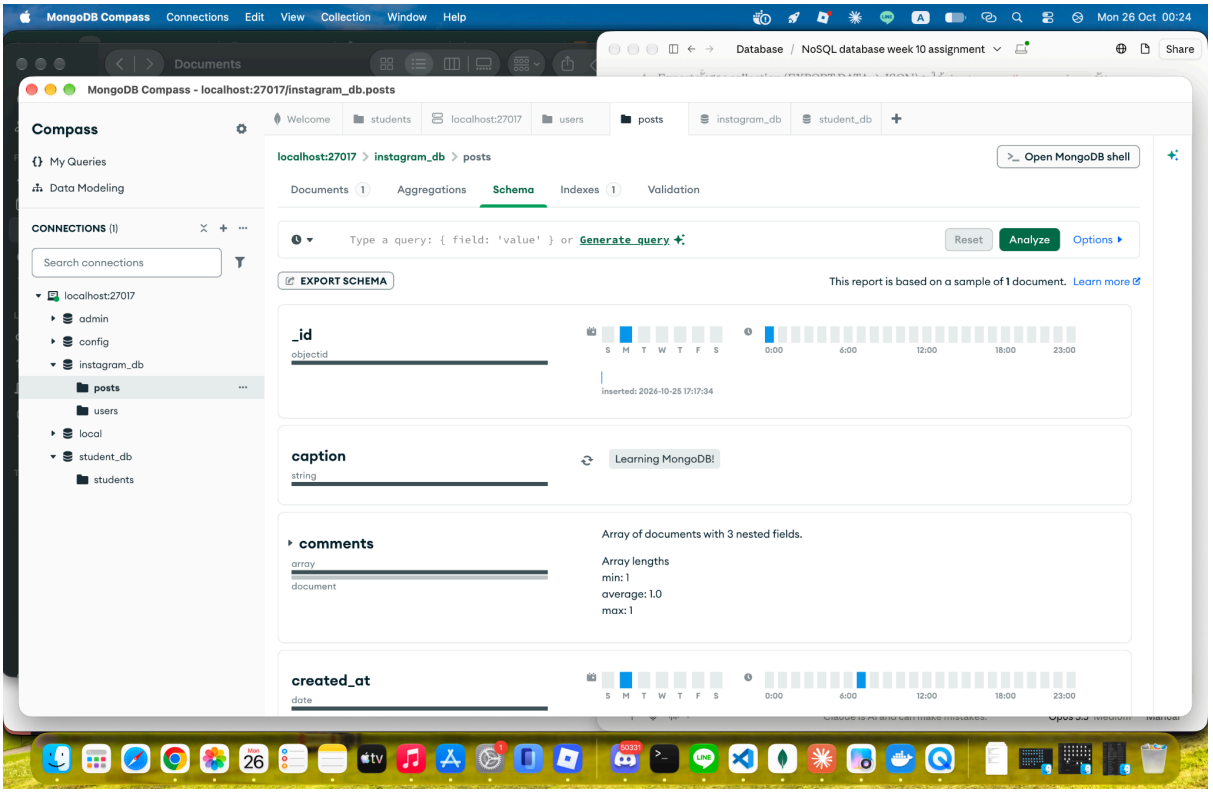
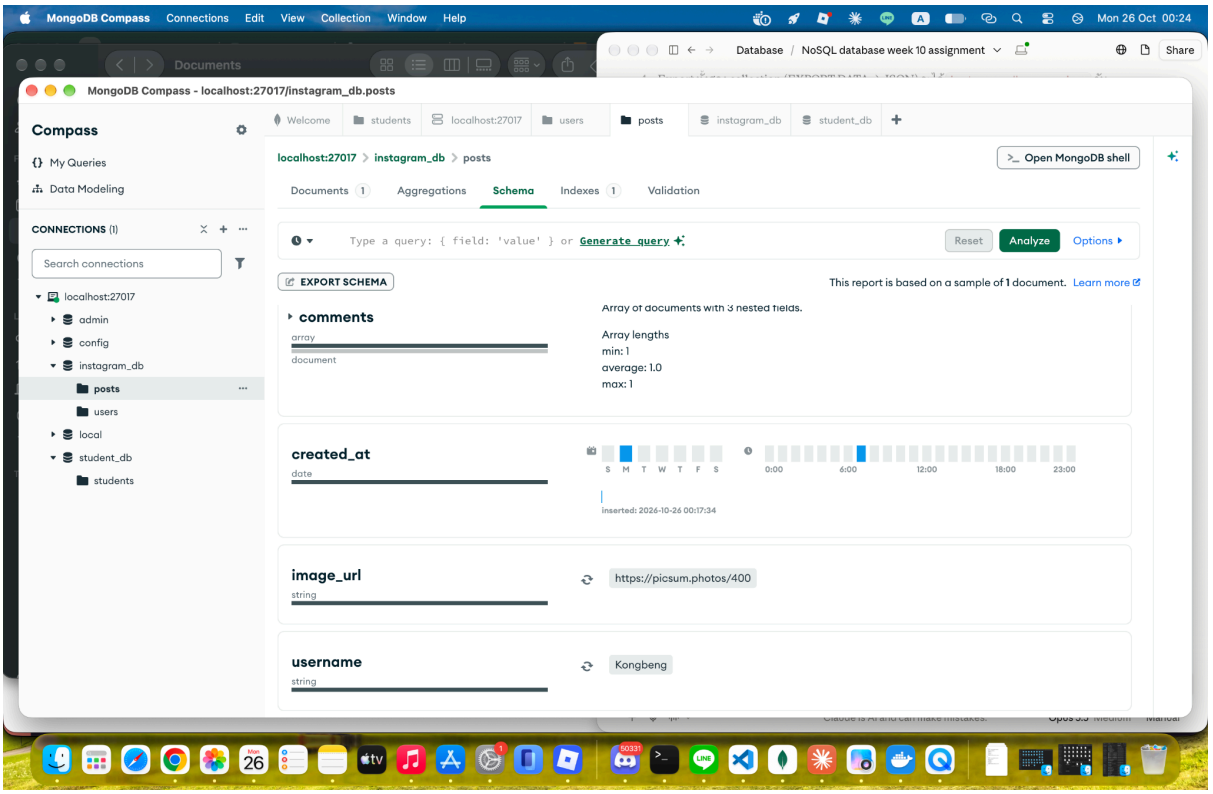

Chrome File Edit View History Bookmarks Profiles Tab Window Help
Mon 26 Oct 00:21
Open WebUI
Algorithm
Week 10:
Screenshot
InstaNoSQL
localhost:5001/feed
School
All Bookmarks
Logged in as: Boom | Feed | My Profile | Logout
Create New Post
Image URL (e.g. https://via.placeholder.com/150)
Caption...
Post
Feed
Kongbeng (Followers: 1) 2026-10-26 00:17
Kongbeng (Followers: 1): Learning MongoDB!
Boom (Followers: 1): Nice
Add a comment...
Comment

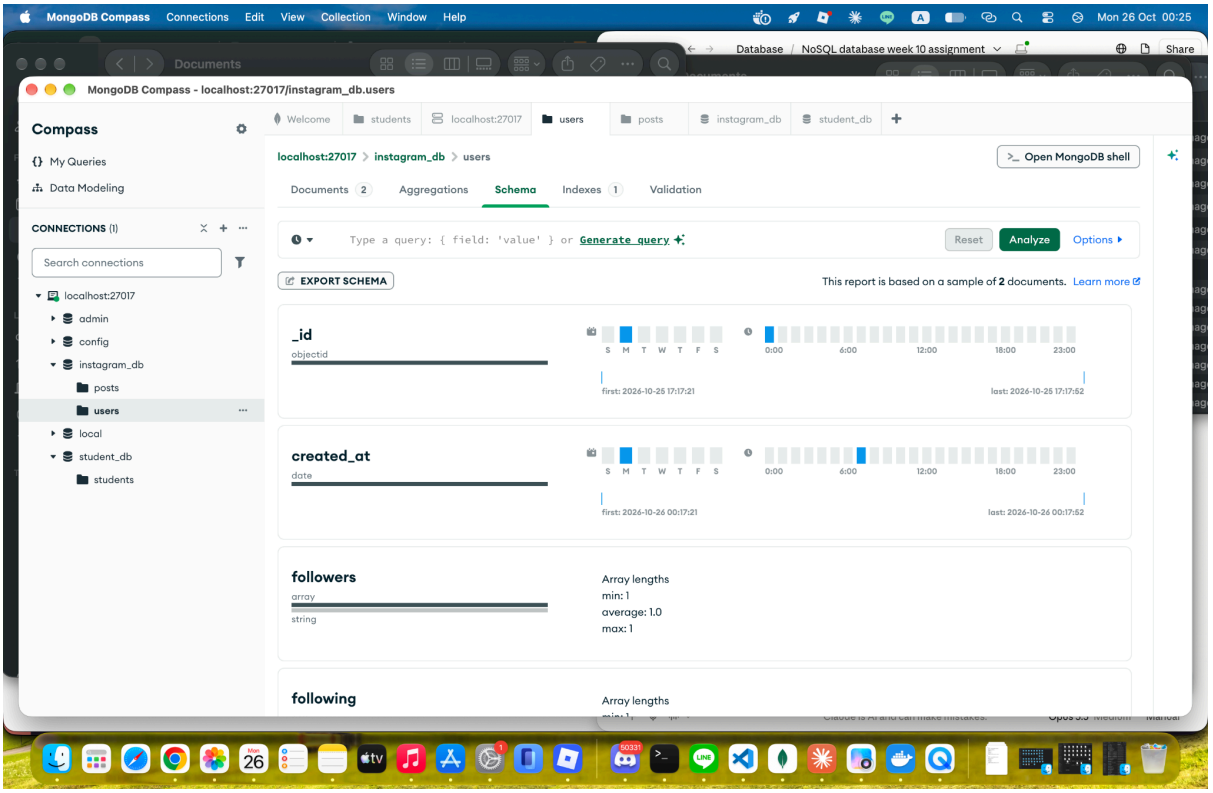

Chrome File Edit View History Bookmarks Profiles Tab Window Help
Mon 26 Oct 00:22
Open WebUI
Algorithm
Week 10:
Screenshot
InstaNoSQL
localhost:5001/user/Kongbeng
School
All Bookmarks
Logged in as: Boom | Feed | My Profile | Logout
Kongbeng's Profile
Followers: 1 | Following: 1
Unfollow
Posts
Learning MongoDB!...
2026-10-26

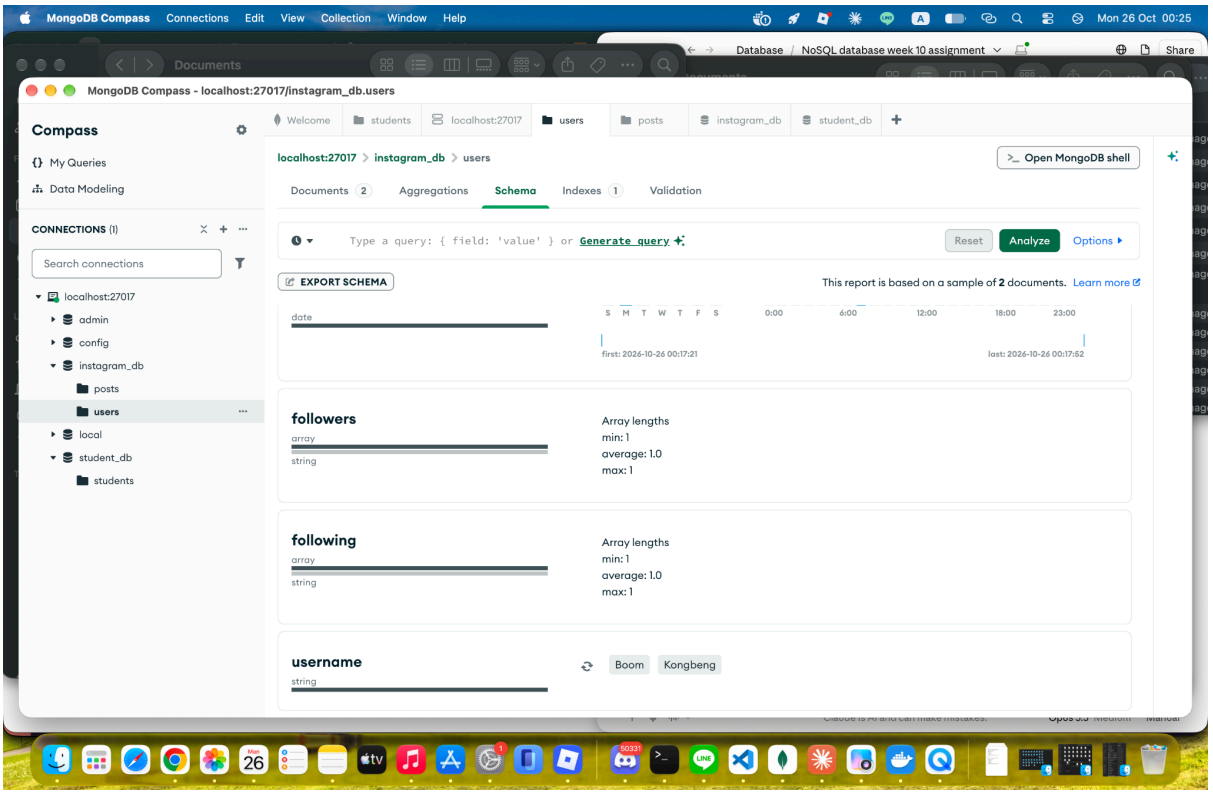

MongoDB Compass - localhost:27017/instagram_db.users
Compass
My Queries
Data Modeling
CONNECTIONS (1)
Search connections
localhost:27017
admin
config
instagram_db
posts
users
local
student_db
students
Welcome
students
localhost:27017
users
student_db
localhost:27017 > instagram_db > users
Open MongoDB shell
Documents 2
Aggregations
Schema
Indexes 1
Validation
Type a query: { field: 'value' } or Generate query
Explain
Reset
Find
Options
ADD DATA
BULK
EXPORT DATA
EXPORT CODE
25
1 - 2 of 2
_id: ObjectId('6ade39a108a40dbfc3b4897a')
username: "Kongbeng"
followers: Array (1)
following: Array (1)
created_at: ISODate('2026-10-26T00:17:21.380+00:00') October 26, 2026 at 12:17:21 AM UTC
_id: ObjectId('6ade39c008a40dbfc3b4897c')
username: "Boom"
followers: Array (1)
following: Array (1)
created_at: ISODate('2026-10-26T00:17:52.776+00:00') October 26, 2026 at 12:17:52 AM UTC


MongoDB Compass - localhost:27017/instagram_db.posts
Compass
My Queries
Data Modeling
CONNECTIONS (1)
Search connections
localhost:27017
admin
config
instagram_db
posts
users
local
student_db
students
Welcome
students
localhost:27017
users
posts
student_db
localhost:27017 > instagram_db > posts
Open MongoDB shell
Documents 1
Aggregations
Schema
Indexes 1
Validation
Type a query: { field: 'value' } or Generate query
Explain
Reset
Find
Options
ADD DATA
BULK
EXPORT DATA
EXPORT CODE
25
1 - 1 of 1
_id: ObjectId('6ade39ae08a40dbfc3b4897b')
username: "Kongbeng"
caption: "Learning MongoDB!"
image_url: "https://picsum.photos/400"
created_at: ISODate('2026-10-26T00:17:34.495+00:00') October 26, 2026 at 12:17:34 AM UTC
comments: Array (1)
0 : Object (3)
username: "Boom"
text: "Nice"
created_at: ISODate('2026-10-26T00:18:02.296+00:00') October 26, 2026 at 12:18:02 AM UTC


MongoDB Compass Connections Edit View Collection Window Help
Mon 26 Oct 00:24
Database / NoSQL database week 10 assignment
Share
MongoDB Compass - localhost:27017/instagram_db.posts
Compass
My Queries
Data Modeling
CONNECTIONS (1)
Search connections
localhost:27017
admin
config
instagram_db
posts
users
local
student_db
students
Welcome
students
localhost:27017
users
posts
instagram_db
student_db
localhost:27017 > instagram_db > posts
Open MongoDB shell
Documents 1
Aggregations
Schema
Indexes 1
Validation
Type a query: { field: 'value' } or Generate query
Reset
Analyze
Options
EXPORT SCHEMA
This report is based on a sample of 1 document. Learn more
_id
objectid
inserted: 2026-10-25 17:17:34
caption
string
Learning MongoDB!
comments
array
document
Array of documents with 3 nested fields.
Array lengths
min: 1
average: 1.0
max: 1
created_at
date


MongoDB Compass Connections Edit View Collection Window Help
Mon 26 Oct 00:24
Database / NoSQL database week 10 assignment
Share
MongoDB Compass - localhost:27017/instagram_db.posts
Compass
My Queries
Data Modeling
CONNECTIONS (1)
Search connections
localhost:27017
admin
config
instagram_db
posts
users
local
student_db
students
Welcome
students
localhost:27017
users
posts
instagram_db
student_db
localhost:27017 > instagram_db > posts
Open MongoDB shell
Documents 1
Aggregations
Schema
Indexes 1
Validation
Type a query: { field: 'value' } or Generate query
Reset
Analyze
Options
EXPORT SCHEMA
This report is based on a sample of 1 document. Learn more
comments
array
document
Array of documents with 3 nested fields.
Array lengths
min: 1
average: 1.0
max: 1
created_at
date
inserted: 2026-10-26 00:17:34
image_url
string
https://picsum.photos/400
username
string
Kongbeng


MongoDB Compass - localhost:27017/instagram_db.users
Compass
My Queries
Data Modeling
CONNECTIONS (1)
Search connections
localhost:27017
admin
config
instagram_db
posts
users
local
student_db
students
Welcome
students
localhost:27017
users
posts
instagram_db
student_db
localhost:27017 > instagram_db > users
Open MongoDB shell
Documents 2
Aggregations
Schema
Indexes 1
Validation
Type a query: { field: 'value' } or Generate query
Reset
Analyze
Options
EXPORT SCHEMA
This report is based on a sample of 2 documents. Learn more
_id
objectid
created_at
date
followers
array
string
Array lengths
min: 1
average: 1.0
max: 1
following


MongoDB Compass - localhost:27017/instagram_db.users
Compass
My Queries
Data Modeling
CONNECTIONS (1)
Search connections
localhost:27017
admin
config
instagram_db
posts
users
local
student_db
students
Welcome
students
localhost:27017
users
posts
instagram_db
student_db
localhost:27017 > instagram_db > users
Open MongoDB shell
Documents 2
Aggregations
Schema
Indexes 1
Validation
Type a query: { field: 'value' } or Generate query
Reset
Analyze
Options
EXPORT SCHEMA
This report is based on a sample of 2 documents. Learn more
date
followers
array
string
Array lengths
min: 1
average: 1.0
max: 1
following
username
string
Boom
Kongbeng

Activity 2 - Task 3: Schema Export & Analysis

1) JSON Structure

Collection: users (2 documents)
- _id : ObjectId
- username : String
- followers : Array of String (usernames)
- following : Array of String (usernames)
- created_at : Date

Collection: posts (1 document)
- _id : ObjectId
- username : String (reference to users.username)
- caption : String
- image_url : String
- created_at : Date
- comments : Array of embedded documents
  - username : String
  - text : String
  - created_at : Date

Modeling choices:
- comments are EMBEDDED in posts (one-to-few, always read together with the post) -> added with $push.
- followers/following are REFERENCES (arrays of usernames) stored in user documents -> updated with $addToSet / $pull on two documents.
- posts reference the author by username; the feed uses $lookup to join posts with users.

2) Equivalent RDBMS Structure

```
CREATE TABLE users (
    user_id    INT AUTO_INCREMENT PRIMARY KEY,
    username   VARCHAR(50) UNIQUE NOT NULL,
    created_at DATETIME
);

CREATE TABLE posts (
    post_id    INT AUTO_INCREMENT PRIMARY KEY,
    user_id    INT NOT NULL,
    caption    TEXT,
    image_url  VARCHAR(255),
    created_at DATETIME,
    FOREIGN KEY (user_id) REFERENCES users(user_id)
);

CREATE TABLE comments (
```

```
    comment_id INT AUTO_INCREMENT PRIMARY KEY,
    post_id    INT NOT NULL,
    user_id    INT NOT NULL,
    text       TEXT,
    created_at DATETIME,
    FOREIGN KEY (post_id) REFERENCES posts(post_id),
    FOREIGN KEY (user_id) REFERENCES users(user_id)
);

CREATE TABLE follows (
    follower_id  INT,
    following_id INT,
    PRIMARY KEY (follower_id, following_id),
    FOREIGN KEY (follower_id)  REFERENCES users(user_id),
    FOREIGN KEY (following_id) REFERENCES users(user_id)
);
```

3) Comparison

- MongoDB: 2 collections. RDBMS: 4 tables (users, posts, comments, follows), because arrays and embedded documents violate 1NF.
- Loading a post with its comments: MongoDB reads 1 document; SQL needs JOIN posts + comments + users.
- Follow: MongoDB updates two user documents with $addToSet (no duplicates). SQL inserts one row into follows; the composite primary key prevents duplicates.
- In MongoDB, followers and following store the same relationship twice (data duplication), so both documents must be updated together. SQL stores it once in the follows table.
- Feed: MongoDB uses an aggregation pipeline ($sort, $lookup, $unwind); SQL uses SELECT ... JOIN users ... ORDER BY created_at.